

## ASTM A516 Carbon Steel, Grade 70

**Categories:** [Metal](#); [Ferrous Metal](#); [Alloy Steel](#); [Low Alloy Steel](#); [ASTM Steel](#); [Carbon Steel](#); [Low Carbon Steel](#)

**Material** Carbon steel plate for moderate and lower temperature service.

**Notes:** Pressure vessel quality

**Key Words:** SA516, ASTM A285

**Vendors:** No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

| Physical Properties | Metric    | English                  | Comments              |
|---------------------|-----------|--------------------------|-----------------------|
| Density             | 7.80 g/cc | 0.282 lb/in <sup>3</sup> | Typical of ASTM Steel |

| Mechanical Properties      | Metric        | English           | Comments             |
|----------------------------|---------------|-------------------|----------------------|
| Tensile Strength, Ultimate | 485 - 620 MPa | 70300 - 89900 psi |                      |
| Tensile Strength, Yield    | 260 MPa       | 37700 psi         |                      |
| Elongation at Break        | 17 %          | 17 %              | in 200mm             |
|                            | 21 %          | 21 %              | in 50mm              |
| Modulus of Elasticity      | 200 GPa       | 29000 ksi         | Typical Carbon Steel |
| Bulk Modulus               | 160 GPa       | 23200 ksi         | Typical for Steel    |
| Poissons Ratio             | 0.29          | 0.29              | Typical Carbon Steel |
| Shear Modulus              | 80.0 GPa      | 11600 ksi         | Typical for Steel    |

| Electrical Properties  | Metric           | English          | Comments             |
|------------------------|------------------|------------------|----------------------|
| Electrical Resistivity | 0.0000170 ohm-cm | 0.0000170 ohm-cm | Typical Carbon Steel |

| Thermal Properties     | Metric       | English                           | Comments             |
|------------------------|--------------|-----------------------------------|----------------------|
| CTE, linear            | 12.0 µm/m-°C | 6.67 µin/in-°F                    | Typical Carbon Steel |
| Specific Heat Capacity | 0.470 J/g-°C | 0.112 BTU/lb-°F                   | Typical Carbon Steel |
| Thermal Conductivity   | 52.0 W/m-K   | 361 BTU-in/hr-ft <sup>2</sup> -°F | Typical Carbon Steel |

| Component Elements Properties | Metric        | English       | Comments                      |
|-------------------------------|---------------|---------------|-------------------------------|
| Carbon, C                     | 0.31 %        | 0.31 %        | may vary with plate thickness |
| Iron, Fe                      | 98.315 %      | 98.315 %      | as balance                    |
| Manganese, Mn                 | 0.85 - 1.2 %  | 0.85 - 1.2 %  |                               |
| Phosphorus, P                 | 0.035 %       | 0.035 %       |                               |
| Silicon, Si                   | 0.15 - 0.40 % | 0.15 - 0.40 % |                               |
| Sulfur, S                     | 0.040 %       | 0.040 %       |                               |

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.